

Evidence-informed bedside teaching card: identify the hemodynamic pattern, give Drug A time to work, pivot at the ceiling, then add Drug B.

RAPID CADENCE: NE q5-10 min | Low-dose epi q10-15 min | Dobutamine q15-20 min | Vasopressin assess over 45-60 min | Milrinone effect evolves over ~6-10 h; longer with renal dysfunction

A. LOW-OUTPUT / STUNNED LV

CI <2.0 | SvO2 <60% | elevated filling pressure | poor perfusion

1 START DRUG A

START

Dobutamine 2-2.5 mcg/kg/min OR epinephrine 0.01-0.02 mcg/kg/min.
Wait 15-20 min between steps.

2 CEILING / PIVOT

PROPOSED

HR >95 or >10-15% above baseline; new ectopy/PVCs; dobutamine reaches 5 mcg/kg/min; or epi reaches 0.04-0.05 mcg/kg/min. Cap Drug A.

3 ADD DRUG B

ADD

Start milrinone 0.125-0.25 mcg/kg/min.
Omit loading bolus in unstable hypotension.
Allow ~6-10 h for steady state; longer in AKI. [1,4]

HOLD / VERIFY WITH MD OR APP

If DBP <50 mmHg or estimated CPP <35 mmHg: hold further escalation, notify MD/APP, and under active order/local protocol add vasopressin 0.02-0.03 U/min and/or rapid NE trim. [1,4]

B. POST-CPB VASOPLEGIA

MAP <65 | SVR <800 | preserved or adequate CI

1 START DRUG A

START

Start norepinephrine 0.02-0.04 mcg/kg/min.
Reassess pressure, flow and perfusion every 5-10 min.

2 CEILING / PIVOT

PROPOSED

NE reaches 0.08-0.10 mcg/kg/min, or SVR >1100 with falling CI: cap NE and add a non-adrenergic mechanism.

3 ADD DRUG B

ADD

Add vasopressin 0.03 U/min under active order/local protocol.
Allow 45-60 min before judging the final response. [1,2]

HOLD / VERIFY WITH MD OR APP

If SVR remains <800 after dual pressors with worsening lactate/acidosis: notify MD/APP.
Provider-only rescue may include methylene blue 2 mg/kg IV over 30 min OR hydroxocobalamin 5 g IV over 15 min after contraindication/interaction review. [1]

C. RV FAILURE / HIGH PVR

Recognize rising CVP, RV dilation/dysfunction or pulmonary hypertension. Protect systemic/diastolic pressure with NE +/- low-dose vasopressin while correcting hypoxia, acidosis, rhythm and preload. Add milrinone 0.125-0.25 mcg/kg/min without bolus and/or an inhaled pulmonary vasodilator under MD/APP order. Preserve DBP/CPP during vasodilation. [1,4]

INTEGRATED RESPONSE - DO NOT TITRATE TO MAP ALONE

After each step reassess: flow (CI/SV/CPO/SvO2), coronary pressure (DBP/CPP), congestion (CVP/PCWP/echo), toxicity (HR, ectopy, ischemia, lactate phenotype), and perfusion (mentation, skin, urine, acid-base trend). Rising dose intensity without recovery should trigger diagnosis/MCS reassessment.

A high pressure can coexist with low stroke volume. Confirm flow, unload the correct vascular compartment, and protect diastolic pressure.

CORE PATTERN: elevated MAP/SVR with low or marginal CI/SV, elevated filling pressure or wall stress. Pressure is not proof of flow. Treat the circulation - not the monitor number alone.

1 VERIFY BEFORE VASODILATION

SAFETY

Confirm adequate preload and review echo/hemodynamics. Exclude pain, agitation, shivering, hypoxemia, tamponade, dynamic/fixed obstruction and severe valve stenosis. Do not vasodilate a hypotensive or underfilled patient. [1,11]

2 CHOOSE THE TARGET

SELECT

High filling pressure, ischemia or arterial-graft/conduit spasm concern: favor nitroglycerin. Predominant arterial afterload excess: favor clevidipine for rapid reversibility or nicardipine for steadier control.

3 START LOW - REASSESS FLOW

DOSE

Under active order/local policy: nitroglycerin 5 mcg/min, increase by 5 mcg/min every 3-5 min initially; OR clevidipine 1-2 mg/h, rapid early titration then every 5-10 min near target; OR nicardipine 5 mg/h, increase by 2.5 mg/h every 5-15 min to a maximum 15 mg/h. [8-10]

NITROGLYCERIN - VENOUS / CORONARY

Best fit: elevated filling pressure, ischemia or arterial-conduit spasm concern. Fast onset/offset. Watch for preload collapse, headache, tachycardia and loss of DBP/CPP. Use non-PVC compatible administration per label/local policy. [1,8]

CLEVIDIPINE - RAPID ARTERIAL

Best fit: arterial afterload excess needing tight, rapidly reversible control. Minimal venous effect. Lipid emulsion: review egg/soy allergy, lipid disorders and daily volume exposure. Watch reflex tachycardia. [10]

NICARDIPINE - STEADY ARTERIAL

Best fit: sustained arterial unloading when a longer control horizon is acceptable. Accumulates more than clevidipine; avoid rapid dose stacking and anticipate a longer hemodynamic tail. [9]

4 IF PRESSURE FALLS BUT FLOW STAYS LOW

PAIR

Recheck preload, rhythm, ischemia and mechanical obstruction. If low contractile reserve persists, pair ordered afterload reduction with dobutamine 2-2.5 mcg/kg/min or milrinone 0.125-0.25 mcg/kg/min without bolus when tachycardia limits catecholamines. [1,4]

5 RESPONSE GATE

REASSESS

Continue only if unloading improves SV/CI/CPO/SvO2 and congestion/perfusion without loss of DBP/CPP. A lower MAP without better forward flow is not success. Avoid stopping an inotrope solely because BP is high until adequate output is documented. [1,11]

HOLD / VERIFY WITH MD OR APP

Hold further reduction and notify MD/APP for falling DBP/CPP, worsening CI/SV/CPO/SvO2, new ischemia/arrhythmia, neurologic change, pulmonary edema, rising filling pressure, worsening urine output, acidosis or lactate. Reassess whether the apparent hypertension is compensatory rather than causal.

BEDSIDER MANTRA: Hypertensive low output is not “too much pressure” alone. It is a mismatch between ventricular reserve and the load being ejected against. The desired response is more forward flow at a safe perfusion pressure.

PK-governed uncoupling: remove toxicity, retain required compensation, wait for offset, then remove the next layer.

ENTRY GATE: stable trend; reversible causes addressed; no active bleeding or mechanical problem. Epi q15-20 min | NE q10-15 min | Vasopressin q45-60 min | Dobutamine q20-30 min | Milrinone q6-8 h; q12-24 h in AKI/CRRT

1 EPINEPHRINE FIRST

TIER 1

0.04 -> 0.02 -> 0.01 -> OFF | q15-20 min.
Why: reduce MVO₂, arrhythmia burden and Type B lactate. [1,3]



2 NOREPINEPHRINE TO LOW BASE

TIER 2

Reduce to 0.02-0.04 mcg/kg/min | q10-15 min.
Retain pressure/inotropic trim while unloading excess adrenergic afterload. [1]



3 VASOPRESSIN NEXT

TIER 3

Once NE <=0.04: 0.03 -> 0.02 -> 0.01 -> OFF | q45-60 min.
Keep low NE as a rapid buffer. Sequence is phenotype-dependent. [5-7]



4 FINISH NOREPINEPHRINE

TIER 4

Residual NE, e.g. 0.02 -> OFF | q10-15 min.
Remember NE has beta-1 contribution; HR and CI may fall when removed. [1]



5 INOTROPES / INODILATOR

TIER 5

Dobutamine 5 -> 2.5 -> 1 -> OFF | q20-30 min.
Milrinone 0.25 -> 0.1875 -> 0.125 -> OFF | q6-8 h; q12-24 h AKI/CRRT. [1,4]

MILRINONE-PRESSOR BRIDGE

- 1 Reduce milrinone only by the ordered step.
- 2 Keep low NE and/or vasopressin to support the vasodilatory tail.
- 3 Wait hours - longer in AKI/CRRT.
- 4 Reassess DBP/CPP + flow + filling + perfusion.
- 5 Then test the pressor after milrinone effect materially recedes. [1,4]

COMBINATION OVERRIDES

Dobutamine + pressor: rapid dobutamine offset permits reassessment in 20-30 min.

NE + vasopressin: no universal drug-off order. Shock cohorts report more hypotension when vasopressin is stopped first; postcardiotomy applicability requires validation. [5-7]

HOLD / VERIFY WITH MD OR APP

Hold further change and notify MD/APP for clinically important falls in DBP/MAP/CPP, CI/SV/CPO/SvO₂; rising CVP/filling pressure; new ischemia or arrhythmia; worsening urine, mentation, skin perfusion, acidosis or lactate. Follow active orders/local policy for return to prior support.

EVIDENCE STATUS

Supported: short catecholamine offset; milrinone long and renal-dependent elimination; milrinone vasodilation may require pressure support; NE has beta-1 activity.
Proposed/local: exact ceiling doses, universal weaning order, fixed vasopressin dose and numeric hold thresholds.

CORE TEACHING POINT: Premature pressor removal during the milrinone tail can create an apparent "failure to wean" that is actually predictable PK mismatch. A normal MAP alone does not prove adequate forward flow.

ABBREVIATIONS

AKI - acute kidney injury
 APP - advanced practice provider
 CI - cardiac index
 CO - cardiac output
 CPO - cardiac power output: $(MAP \times CO) / 451$
 CPP - coronary perfusion pressure; approximated as aortic DBP minus LVEDP
 CRRT - continuous renal replacement therapy
 CVP - central venous pressure
 DBP / AoDP - diastolic / aortic diastolic pressure
 Epi - epinephrine
 HR - heart rate
 LV / RV - left / right ventricle
 LVEDP - left ventricular end-diastolic pressure
 MAP - mean arterial pressure
 MCS - mechanical circulatory support
 MVO₂ - myocardial oxygen consumption
 NE - norepinephrine
 PAC - pulmonary artery catheter
 PCWP - pulmonary capillary wedge pressure
 PK - pharmacokinetics
 PVR / SVR - pulmonary / systemic vascular resistance
 SvO₂ - mixed venous oxygen saturation

EVIDENCE & CERTAINTY KEY

Established pharmacology

Short catecholamine offset; milrinone vasodilation, renal elimination, ~2.3-2.4 h terminal half-life and ~6-12 h to steady-state concentration. [4]

Evidence-informed clinical practice

Use complementary mechanisms to limit catecholamine toxicity; vasopressin is a studied alternative in postcardiac-surgery vasoplegia. [1,2]

Proposed pilot thresholds

Exact "low-reserve" ceiling doses, fixed titration intervals, numeric hold thresholds, and the full universal sequence have not been validated as one protocol. They must be evaluated against the patient's hemodynamic pattern, active orders and local scope.

Important inference

Maintaining pressure support during the residual milrinone effect is pharmacologically coherent, but the specific bridge sequence requires prospective bedside validation.

SELECTED REFERENCES

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PILOT PURPOSE: Evaluate clarity, bedside concordance, perceived safety, missing scenarios, and whether PK-aware sequencing improves clinicians' ability to explain escalation or apparent "failure to wean." This card is not authorization for independent medication changes.